

Exploratory Data Analysis (EDA) Summary Report Template

Task 1 — Exploratory Data Analysis and Key Risk Indicators

1. Introduction

Purpose: assess the structure, completeness, consistency, and exploratory risk signals in Geldium's customer delinquency dataset before predictive modeling. The analysis is intended to identify data-quality issues, candidate risk indicators, and modeling considerations while protecting customer confidentiality.

2. Dataset Overview

This section summarizes the dataset, including the number of records, key variables, and data types. It also highlights any anomalies, duplicates, or inconsistencies observed during the initial review.

Key dataset attributes:

- Number of records: 500 customers across 19 variables.
- Key variables: Customer_ID; Age; Income; Credit_Score; Credit_Utilization; Missed_Payments; Delinquent_Account; Loan_Balance; Debt_to_Income_Ratio; Employment_Status; Account_Tenure; Credit_Card_Type; Location; and six monthly payment-status fields (Month_1 to Month_6).
- Data types: 9 numerical variables and 10 categorical/text variables (including Customer_ID). No duplicate rows were found.

3. Missing Data Analysis

Identifying and addressing missing data is critical to ensuring model accuracy. This section outlines missing values in the dataset, the approach taken to handle them, and justifications for the chosen method.

Key missing data findings:

- Variables with missing values: Income — 39 (7.8%); Loan_Balance — 29 (5.8%); Credit_Score — 2 (0.4%). Total missing cells: 70 of 9,500 (0.74%).
- Missing-data treatment: retain the original source data unchanged; for modeling, use median imputation for numerical fields (preferably calculated on the training set only) and add missingness indicators where appropriate. Avoid deleting rows because the missingness is concentrated in only three fields and row deletion would unnecessarily reduce the sample.

4. Key Findings and Risk Indicators

This section identifies trends and patterns that may indicate risk factors for delinquency. Feature relationships and statistical correlations are explored to uncover insights relevant to predictive modeling.

Key findings:

- Overall delinquency prevalence: 16.0% (80 of 500 customers). Simple correlations with the delinquency flag are weak, so correlation alone should not be used to rank predictors. Exploratory group

comparisons show slightly higher average Credit Utilization among delinquent customers (50.7% vs 48.9%) and slightly higher Debt-to-Income Ratio (30.6% vs 29.7%).

- Key anomalies requiring investigation: 4 records (0.8%) have Credit_Utilization above 100%, which should be checked against the source-system definition. Employment_Status contains inconsistent capitalization/abbreviation variants and should be standardized. Missing Income, Loan_Balance, and Credit_Score values should be handled using a documented training-set-only imputation process.

5. AI & GenAI Usage

Generative AI tools were used to summarize the dataset, impute missing data, and detect patterns. This section documents AI-generated insights and the prompts used to obtain results.

Example AI prompts used:

- 'Summarize key patterns in the dataset and identify anomalies.'
- 'Suggest an imputation strategy for missing income values based on industry best practices.'
- Top 3 candidate variables for further delinquency modeling: 1) Credit Utilization — slightly higher among delinquent customers and may indicate greater credit pressure; 2) Debt-to-Income Ratio — slightly higher among delinquent customers and may reflect repayment burden; 3) Credit Score — an important candidate risk feature, but its predictive strength should be validated through modeling rather than assumed from simple correlation.
- GenAI-assisted prompts used: (1) “Summarize key patterns in the dataset and identify anomalies.” (2) “Suggest an imputation strategy for missing income values based on industry best practices.” (3) “Identify variables that may be predictive of delinquency without exposing individual customer information.” (4) “Review categorical fields for inconsistent labels and propose a standardization approach.” AI outputs should be validated against source data and should not receive raw personally identifiable or unnecessary financial information.

Variable		Missing		Missing %	
Income		39		7.8%	
Credit_Score		2		0.4%	
Loan_Balance		29		5.8%	
Month	On-time		Late		Missed
Month 1	177		159		164
Month 2	160		173		167
Month 3	169		169		162
Month 4	159		181		160
Month 5	162		151		187
Month 6	160		172		168

Illustrative exploratory segment rates (not causal evidence):

Segment	Category	Delinquency Rate
Employment	Unemployed	19.4%
Employment	Employed	18.3%
Employment	Self-employed	16.2%
Card type	Business	21.3%
Card type	Student	17.9%
Card type	Gold	14.4%
Location	Los Angeles	19.6%
Location	Houston	16.8%
Location	Phoenix	16.5%

Methodology note: Findings are descriptive and based only on the supplied dataset. Segment differences should be validated on a larger, representative sample and against business definitions before being used for collections decisions.

6. Conclusion & Recommendations

Conclusion

The EDA shows that the dataset contains 500 customer records and 19 variables, with a delinquency rate of 16%. The data is generally usable, but missing values in Income, Loan Balance, and Credit Score, inconsistent Employment Status labels, and four Credit Utilization values above 100% require attention before predictive modeling. Credit Utilization, Debt-to-Income Ratio, and Credit Score are the three key candidate risk indicators identified for further investigation.

Recommendations

- Standardize inconsistent categorical values, particularly Employment Status.
- Handle missing numerical values using a documented imputation strategy and retain missingness indicators where appropriate.
- Validate Credit Utilization values above 100% against the source-system definition.
- Use Credit Utilization, Debt-to-Income Ratio, and Credit Score as candidate features for the next modeling stage.
- Define the prediction window and target carefully to prevent data leakage.
- Evaluate models using discrimination, calibration, and fairness across relevant customer groups before deployment.
- Use model outputs to prioritize appropriate collections interventions rather than making automated decisions solely from the risk score.

Next Steps

Proceed to feature preparation and predictive-model development after resolving the identified data-quality issues. The modeling stage should compare interpretable and nonlinear approaches and assess predictive performance, calibration, and fairness before any operational collections use.